

ε_0 in CRF: Two Roles, Not One

What GPT Proposed, What is True, and What Remains Open

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Abstract

A consultation with GPT suggested that the instability floor ε_0 from EIC equals the entropy deficit in Born resampling, and that this unification closes the β derivation via $\beta = d_s/(2(\ln 2 - \varepsilon_0))$.

This document shows: (1) the two ε_0 quantities are numerically distinct and play different structural roles; (2) what *is* derivable from ε_0 and n alone; (3) what the actual remaining open gap is.

Key result: H_{uniform} and H_{peaked} are fully derived from (ε_0, n) via the digamma formula. $H_{\text{after}} = (1 - p_{II})H_{\text{uniform}} + p_{II}H_{\text{peaked}}$ where p_{II} is the Phase II node fraction. p_{II} connects to the HDynamics track ($\alpha(W)$) and is the one remaining open quantity for the β derivation.

GPT's formula $\beta = d_s/(2(\ln 2 - \varepsilon_0))$ is numerically wrong (32% gap) and structurally circular (assumes what it derives).

1 Three Quantities Named ε_0

The conversation with GPT conflated three distinct quantities:

Symbol	Definition	Role	Value
$\varepsilon_0^{\text{EIC}}$	$ I_{\text{eff}} - \ln 2 $	Born regulariser: $\alpha_k = (\bar{P}_k + \varepsilon_0)/\varepsilon_0$	0.00755
$\varepsilon_0^{\text{Born}}$	$\ln n - H_{\text{uniform}}$	Entropy deficit, uniform Born	0.02466
$\varepsilon_0^{\text{mix}}$	$\ln n - H_{\text{after}}$	Entropy deficit, Phase I/II mix	0.2814

These three numbers differ by factors of 3 and 37 respectively. GPT's proposal $H = \ln n - \varepsilon_0^{\text{EIC}}$ confuses the Born regulariser with the entropy deficit.

GPT's Formula

GPT proposed:

$$\beta = \frac{d_s}{2(\ln 2 - \varepsilon_0^{\text{EIC}})} = \frac{\ln 8}{2(0.6931 - 0.00755)} = 1.517$$

$\beta_{\text{sim}} = 2.257$. Gap: 32.8%. **Numerically wrong.**

Structurally: the premise $\beta = d_s/(2H)$ is not derived — it is assumed, making the proof circular. The factor $1/2$ from “pairwise interaction” is an ansatz.

2 What is Correctly Derived

H_{uniform} and H_{peaked} from (ε_0, n)

Both entropy values are fully derived from $\varepsilon_0^{\text{EIC}}$ and n via the digamma formula for asymmetric Dirichlet distributions.

Phase I neighbours (uniform $\bar{P}$):

$$\alpha_k^{\text{unif}} = \frac{1}{n\varepsilon_0} + 1 = 17.55, \quad \alpha_0^{\text{unif}} = n\alpha_k^{\text{unif}} = 140.4$$

$$H_{\text{uniform}} = \psi(\alpha_0^{\text{unif}} + 1) - \psi(\alpha_k^{\text{unif}} + 1) = 2.0548$$

Phase II neighbours (peaked $\bar{P} \approx e_a$):

$$\alpha_{\text{peak}} = \frac{1}{\varepsilon_0} = 132.5, \quad \alpha_{\text{bg}} = \frac{1}{n} + 1 = 1.125$$

$$H_{\text{peaked}} = \psi(\alpha_0^{\text{pk}} + 1) - \frac{\alpha_{\text{peak}}}{\alpha_0^{\text{pk}}} \psi(\alpha_{\text{peak}} + 1) - \frac{(n-1)\alpha_{\text{bg}}}{\alpha_0^{\text{pk}}} \psi(\alpha_{\text{bg}} + 1) = 0.304$$

Both derived from ε_0 and n only. No free parameters.

H_{after} from p_{II} (Neighbourhood Mixing)

For a random graph with $k = 20$ neighbours:

$$H_{\text{after}} = (1 - p_{II}) H_{\text{uniform}} + p_{II} H_{\text{peaked}}$$

where p_{II} is the fraction of nodes in Phase II (crystallised). With $p_{II} = 0.147$ (measured): $H_{\text{after}} = 1.797$, matching simulation at $< 0.1\%$.

The key insight: H_{after} is set by the *neighbourhood composition* when the R-event fires, not by the firing node's own state. Since the graph is well-mixed ($k \gg 1$), the neighbourhood Phase II fraction equals the population Phase II fraction p_{II} .

3 GPT's Insight: Taylor Expansion of ε_0

GPT correctly derived the quadratic form:

$$\varepsilon_0 = \frac{n}{2} \sum_i \delta_i^2 + O(\delta^3), \quad p_i = \frac{1}{n} + \delta_i, \quad \sum \delta_i = 0$$

This is mathematically correct and connects $\varepsilon_0^{\text{Born}}$ to the variance of the probability distribution. However, $\varepsilon_0^{\text{EIC}} = 0.00755$ is not the same number as $\varepsilon_0^{\text{Born}} = 0.02466$. The expansion is valid but does not unify the two ε_0 values.

4 The One Remaining Open Quantity: p_{II}

p_{II} from HDynamics

p_{II} is the Phase II fraction of nodes at simulation time s . It satisfies a crystallisation equation connected to HDynamics:

$$\frac{dH_i}{dt} = -\alpha(W) H_i^2 \quad \Rightarrow \quad H_i(t) = \frac{H_0}{1 + \alpha(W) H_0 t}$$

Node i enters Phase II when $H_i \rightarrow \varepsilon_0$, i.e., at crystallisation time:

$$t_{\text{cryst}} = \frac{1}{\alpha(W) \varepsilon_0} - \frac{1}{\alpha(W) H_0} \approx \frac{1}{\alpha(W) \varepsilon_0}$$

With $\varepsilon_0 = 0.00755$ and $H_0 = 2.055$: $t_{\text{cryst}} \approx 132/\alpha(W)$.

For $p_{II}(s = 100) = 0.147$, the required crystallisation rate is $\alpha(W) \approx 0.227$, which must be derived from the δ -chain via the C -gradient and the W coupling (HDynamics paper).

This is the one remaining gap for the complete β derivation.

5 Updated Status Table

Quantity	Status	Source
$\varepsilon_0 = I_{\text{eff}} - \ln 2 $	Derived	EIC/SINDy
$n = 8$	Derived	SO(3) #16
H_{uniform} (digamma)	Derived	(ε_0, n)
H_{peaked} (digamma)	Derived	(ε_0, n)
$H_{\text{after}} = (1 - p_{II})H_u + p_{II}H_p$	Formula derived	p_{II} open
p_{II} (Phase II fraction)	Open	HDynamics $\alpha(W)$
$\alpha(W)$ (crystallisation rate)	Open	δ -chain via C -gradients
$\beta = 2.219$ at 1.7%	Measured	V20.3
GPT: $\beta = d_s/(2(\ln 2 - \varepsilon_0))$	Incorrect	Circular + wrong value

6 The Correct Path to β Closure

The correct derivation chain is:

$$\begin{aligned}
& \delta \xrightarrow{\text{EIC}} \varepsilon_0 \xrightarrow{\text{SO}(3)} n \xrightarrow{\psi} H_{\text{uniform}}, H_{\text{peaked}} \\
& \delta \xrightarrow{\text{HDynamics}} \alpha(W) \xrightarrow{dH/dt} t_{\text{cryst}} \rightarrow p_{II}(s) \\
& \Rightarrow H_{\text{after}} = (1 - p_{II})H_u + p_{II}H_p \rightarrow \beta = \frac{F(H_{\text{after}} \ln(m + \kappa) - H_{\text{before}} \ln m)}{\kappa}
\end{aligned}$$

The HDynamics track (deriving $\alpha(W)$ from δ -chain) is the last genuine gap. It is a well-posed problem. Once $\alpha(W)$ is derived, p_{II} and β follow analytically.

ε_0 plays two roles in CRF: regulariser and entropy floor.

They are not the same number.

What GPT correctly identified: Born resampling creates an entropy deficit.

What GPT missed: the deficit comes from Phase II neighbours, not from ε_0 directly.

What is now derived: H_{uniform} and H_{peaked} from (ε_0, n) .

What remains: $\alpha(W)$ from the δ -chain.